Clearance & Elimination Practice 4

Module 4 - Drug Elimination and Clearance · Daily practice sheet · No date

Name _____ Date _____

1a 1. A new antibiotic is excreted by the kidney. The apparent volume of distribution is 25L in the normal adult. The clearance of this drug is 300 mL/min. a. What is the probable mechanism of renal excretion of this drug?

A. Hepatic metabolism

B. Filtration with active secretion

C. Filtration with tubular reabsorption

D. Filtration only

1b 1. A new antibiotic is excreted by the kidney. The apparent volume of distribution is 25L in the normal adult. The clearance of this drug is 300 mL/min. b. What is the usual t½ for this drug?

1c 1. A new antibiotic is excreted by the kidney. The apparent volume of distribution is 25L in the normal adult. The clearance of this drug is 300 mL/min. c. What would be the new t½ for this drug in an adult with partial renal failure whose clearance of the antibiotic was 125 mL/min?

Problem 2 Estimate the CrCl of a 42-year-old female who weighs 82kg, S_{Cr} =0.9 mg/dL, and 170 cm tall.

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

1a

Her key

Answer: B. Filtration with active secretion

WHAT THE STEM GIVES

V_D	25L	*not needed for the mechanism; used for the half-life in part b*
Cl_T	300 mL/min	equals Cl_R, since excretion is renal; compared with about 120 mL/min

THE WORKING, ONE LINE AT A TIME

SETUP

The drug is excreted by the kidney, so the 300 mL/min clearance is its renal clearance; reference GFR about 120 mL/min

Nothing else eliminates the drug, so total and renal clearance are the same number, read against the glomerular filtration rate of about 120 mL/min.

ALGEBRA

$$\frac{300 \text{ mL/min}}{120 \text{ mL/min}} = 2.5;\ 300 - 120 = 180 \text{ mL/min beyond filtration}$$

Each minute 180 mL of plasma more than filtration can account for is cleared of the drug, so a second process is adding drug to the tubular fluid.

SETUP

Compare: Cl_R = 300 mL/min is above 120 mL/min, so filtration with active tubular secretion

Above the GFR means a process adds drug to the lumen beyond filtration: active secretion. Reabsorption could only pull the value below 120 mL/min.

> **DOES THE NUMBER MAKE SENSE?**
>
> Reabsorption can only lower renal clearance below the filtration rate, and filtration alone cannot exceed about 120 mL/min, so 300 mL/min, 2.5 times the GFR, can come from nothing but filtration plus active secretion. The 25 L volume does not enter.

Clearance, 300 mL/min, is well above the GFR, the glomerular filtration rate of about 120 mL/min. Filtration cannot clear more plasma than it filters, so a transporter must add drug to the tubule: filtration with active secretion.

1b

Her key

Answer: 58 min

WHAT THE STEM GIVES

route	excreted by the kidney	*not needed for the number*
V_D	25L	put in mL, 25,000 mL, the numerator of 0.693 V_D
Cl	300 mL/min	the denominator

THE WORKING, ONE LINE AT A TIME

UNIT

$V_D = 25\ L = 25{,}000\ mL$

Clearance is given in mL/min, so the volume is converted to mL as well; then mL cancels and the half-life comes out in minutes.

ALGEBRA

$$t½ = \frac{(0.693)(25{,}000\ mL)}{300\ mL/min} = 58\ min$$

(0.693)(25,000 mL) = 17,325 mL on top; dividing by Cl = 300 mL/min, the millilitres cancel and the reciprocal minutes invert to minutes: $\frac{17{,}325\ mL}{300\ mL/min} = 57.75$ min, reported as 58 min. Half of the drug in the body is cleared every 58 minutes.

> **DOES THE NUMBER MAKE SENSE?**
>
> Cl of 300 mL/min empties the 25,000 mL once in $\frac{25{,}000}{300}$ min, and a half-life is 0.693 of that turnover time, so a little under an hour: 58 min fits. The unit is minutes because Cl is per minute.

1c Her key

Answer: 139 min

WHAT THE STEM GIVES

route	excreted by the kidney	*not needed for the number*
V_D	25L	unchanged: 25,000 mL, the numerator of 0.693 V_D
Cl	300 mL/min	*not needed: replaced by the lower clearance*
Cl new	125 mL/min	the new denominator

THE WORKING, ONE LINE AT A TIME

ALGEBRA

$$t½ = \frac{(0.693)(25{,}000\ mL)}{125\ mL/min} = 139\ min$$

$t½ = \frac{0.693\ V_D}{Cl}$: V_D, the volume of distribution, is unchanged, so (0.693)(25,000 mL) = 17,325 mL on top, and with Cl lowered to 125 mL/min, $\frac{17{,}325\ mL}{125\ mL/min} = 138.6$ min. Clearance fell by the factor $\frac{300}{125} = 2.4$, so the half-life rose 2.4 times, from 58 to 139 min.

> **DOES THE NUMBER MAKE SENSE?**
>
> Clearance fell from 300 to 125 mL/min, less than half, so the half-life more than doubles from the part b value; 139 min fits, and a value shorter than part b means the ratio was inverted.

Problem 2 Her key

Answer: 79.19 mL/min

WHAT THE STEM GIVES

age	42-year-old	140 − 42 = 98 on top
sex	female	IBW starts at 45.5 kg and the result is multiplied by 0.85
weight	82kg	*not needed: her keys use ideal body weight*
S_{Cr}	0.9 mg/dL	the denominator, 72 × 0.9 = 64.8
height	170 cm	$\frac{170}{2.54}$ = 66.93 in, 7 in over 5 ft, so IBW = 61.6 kg

THE WORKING, ONE LINE AT A TIME

UNIT

170 cm ÷ 2.54 = 66.93 in, taken as 7 in over 5 ft

Height in centimetres is divided by 2.54 cm per inch: $\frac{170\ \text{cm}}{2.54\ \text{cm/in}}$ = 66.93 in, the centimetres cancelling. The ideal body weight equation counts inches above 60 in (5 ft), so 66.93 − 60 = 6.93, rounded to a whole 7 in before the IBW is computed, as her keys do.

ALGEBRA

$$\text{IBW} = 45.5 + 2.3(7) = 61.6\ \text{kg}$$

The female form starts at 45.5 kg. The 82 kg total body weight is not used, because her keys use ideal body weight.

SETUP

$$\text{CrCl} = 0.85 \times \frac{(140 - 42)(61.6)}{(72)(0.9)} = 79.19\ \text{mL/min}$$

The Cockcroft-Gault equation estimates creatinine clearance from age, ideal body weight and serum creatinine S_{Cr}: (98)(61.6) = 6036.8 over (72)(0.9) = 64.8 gives 93.16 mL/min; the patient is female, and women produce less creatinine per kilogram, so 0.85 × 93.16 = 79.19 mL/min.

> **DOES THE NUMBER MAKE SENSE?**
> With IBW 61.6 kg the male estimate is 140 − 42 times 61.6 over 72 × 0.9, a little over 90 mL/min; 0.85 of that is near 79 mL/min, so 79.19 fits. Skipping the 0.85 gives a value too high by about a sixth.

With 6.93 inches instead of 7, IBW is 61.44 kg and CrCl 78.98 mL/min; both are accepted.